AP PRECALCULUS · TOPIC 1.11

Equivalent Representations of Polynomial and Rational Expressions

Section 1 — Multiple Choice
Circle the letter of the correct answer **SOLVED**

Name ..... Class _____ Date _____

No credit for unsupported answers in Section 2. Questions 1–8: 1 point each (8 points).

1 Let $f(x) = x^3 - 4x^2 - 5x$. Which list gives all of the zeros of f?

A. $-1,\ 0,\ 5$ **B.** $-5,\ 0,\ 1$

C. $-1,\ 5$ **D.** $0,\ 4,\ 5$

A. $f(x) = x(x-5)(x+1)$. B: sign error, reading each factor $x - a$ as $x = -a$. C: divided out the common factor x and lost the zero at 0. D: read the coefficients 4 and 5 as zeros.

2 Let $r(x) = \dfrac{x^2 - x - 6}{x^2 - 9}$. What are the coordinates of the hole in the graph of r?

A. $(3, 0)$ **B.** $(3, 1)$

C. $\left(3, \dfrac{5}{6}\right)$ **D.** $\left(3, \dfrac{6}{5}\right)$

C. $r(x) = \dfrac{(x-3)(x+2)}{(x-3)(x+3)}$, so the hole is where $x = 3$ and the reduced value is $\dfrac{3+2}{3+3} = \dfrac{5}{6}$. A: used the original numerator, which is 0 at $x = 3$. B: used the horizontal asymptote as the y-value. D: inverted the reduced fraction.

3 What is the quotient when $3x^3 - 2x^2 + x - 4$ is divided by $x - 1$?

A. $3x^2 - 2x + 1$ **B.** $3x^2 + x + 2$

C. $3x^2 - 5x + 6$ **D.** $3x^2 + x$

B. Synthetic division with 1: $3, -2, 1, -4 \to 3, 1, 2, -2$; quotient $3x^2 + x + 2$, remainder -2. A: copied the coefficients without adding the carried values. C: divided by $x + 1$ (used -1). D: added the remainder -2 into the quotient.

4 Which line is the slant asymptote of $r(x) = \dfrac{x^2 + 5x + 3}{x + 2}$?

A. $y = x + 5$ **B.** $y = x - 3$

C. $y = x + 2$ **D.** $y = x + 3$

D. Synthetic division with -2: $1, 5, 3 \to 1, 3, -3$; $r(x) = x + 3 - \dfrac{3}{x+2}$. A: used the numerator's middle coefficient. B: used the remainder as the constant. C: used the divisor's constant.

5 What is the coefficient of x^2 in the expansion of $(x - 2)^4$?

A. -24 **B.** 6

C. 12 **D.** 24

D. Row 4 is $1, 4, 6, 4, 1$, so the term is $6 \cdot x^2 \cdot (-2)^2 = 24x^2$. A: made the sign negative (even power of -2 is positive). B: used the Pascal number only. C: used $6 \cdot 2$, forgetting to square -2.

6 Which statement about the graph of $r(x) = \dfrac{x^2 - 4}{x^2 - x - 2}$ is true?

A. Vertical asymptotes at $x = 2$ and $x = -1$

B. A hole at $x = 2$ and a vertical asymptote at $x = -1$

C. A hole at $x = -1$ and a vertical asymptote at $x = 2$

D. A hole at $x = -2$ and a vertical asymptote at $x = 1$

B. $r(x) = \dfrac{(x-2)(x+2)}{(x-2)(x+1)}$: the factor $x - 2$ cancels (hole), $x + 1$ remains (asymptote). A: treated every zero of the denominator as an asymptote. C: swapped the roles of the two factors. D: sign error, reading $x - a$ as $x = -a$.

7 Which statement describes the end behavior of $f(x) = -2(x+1)^2(x-3)$?

A. $f(x) \to \infty$ as $x \to \infty$ and $f(x) \to -\infty$ as $x \to -\infty$

B. $f(x) \to -\infty$ as $x \to \infty$ and as $x \to -\infty$

C. $f(x) \to -\infty$ as $x \to \infty$ and $f(x) \to \infty$ as $x \to -\infty$

D. $f(x) \to \infty$ as $x \to \infty$ and as $x \to -\infty$

C. Leading term $-2x^2 \cdot x = -2x^3$: odd degree, negative coefficient. A: ignored the negative leading coefficient. B: treated the squared factor as making the whole degree even. D: both errors together.

8 Which is the expansion of $(2x-1)^3$?

A. $8x^3 - 12x^2 + 6x - 1$

B. $2x^3 - 3x^2 + 3x - 1$

C. $8x^3 + 12x^2 + 6x + 1$

D. $8x^3 - 1$

A. Row 3 is $1, 3, 3, 1$ with $a = 2x$, $b = -1$. B: did not cube the 2 in $2x$. C: dropped the negative signs. D: used $(a-b)^3 = a^3 - b^3$, losing the middle terms.

AP PRECALCULUS · TOPIC 1.11 · SECTION 2

Equivalent Representations of Polynomial and Rational Expressions

Section 2 — Free Response
Show all working; answers need units and limit notation where relevant

Question 9 Calculator allowed · 6 points

A drone-delivery company models its average cost per delivery, in dollars, by $A(n) = \dfrac{n^2 + 10n + 40}{n + 5}$, where $n \geq 1$ is the number of hundreds of deliveries made in a month.

(a) Use polynomial long division to write $A(n)$ as a quotient plus a remainder over $n + 5$. DOK3 2 pt

Synthetic division with -5: $1, 10, 40 \to 1, 5, 15$. $A(n) = n + 5 + \dfrac{15}{n+5}$.

(b) Write the equation of the slant asymptote of the graph of A and interpret it in context. DOK3 2 pt

$y = n + 5$. For large monthly volumes the average cost per delivery is close to $n + 5$ dollars, which keeps growing with n in this model.

(c) Find the number of hundreds of deliveries for which the average cost is exactly 20 dollars. Give your answer to three decimal places. DOK3 2 pt

$n^2 + 10n + 40 = 20n + 100 \Rightarrow n^2 - 10n - 60 = 0 \Rightarrow n = 5 + \sqrt{85} \approx 14.220$ hundred deliveries.

Question 10 No calculator · 6 points

Let $p(x) = x^3 - 3x^2 - 4x + 12$ and $r(x) = \dfrac{p(x)}{(x-2)(x+1)}$.

(a) Factor $p(x)$ completely and list its zeros. DOK3 2 pt

$p(x) = x^2(x-3) - 4(x-3) = (x-3)(x-2)(x+2)$; zeros $-2,\ 2,\ 3$.

(b) Write $r(x)$ in simplified form and give the coordinates of the hole. DOK3 2 pt

$r(x) = \dfrac{(x-3)(x+2)}{x+1}$. At $x = 2$: $\dfrac{(-1)(4)}{3} = -\dfrac{4}{3}$. Hole at $\left(2, -\dfrac{4}{3}\right)$.

(c) Use long division on the simplified form to find the slant asymptote of r. DOK3 2 pt

$\dfrac{x^2 - x - 6}{x+1}$: synthetic division with -1: $1, -1, -6 \to 1, -2, -4$. Slant asymptote $y = x - 2$.

Answer key (Section 1): 1A 2C 3B 4D 5D 6B 7C 8A